

Department of Information & Communication Technology

Faculty of Technology

Rajarata University of Sri Lanka

ICT 3310 – Information Security

Assignment (Written)

Assignment Title:

Evaluation of Security Measures Implemented in WhatsApp

Objective:

- The purpose of this assignment is to assess students' ability to analyze, evaluate, and present findings on the security mechanisms used in a widely adopted communication platform.
- Students will investigate how WhatsApp protects user data, prevents unauthorized access, and maintains privacy, as well as identify potential weaknesses.

Instructions

Report Length: 2,000 – 3,000 words (excluding references and appendices).

Structure: Your report should follow a formal academic style with the following sections:

1. Cover Page
 - Title of the report
 - Student name, Registration No., Index No.
 - Course name and code
 - Date of submission
2. Abstract (250–300 words)
 - A concise summary of your research, findings, and conclusion.
3. Introduction
 - Background of WhatsApp and its global usage.
 - Importance of security in instant messaging platforms.
 - Purpose and scope of the report.
4. Overview of WhatsApp Security Architecture
 - End-to-end encryption mechanism.
 - Authentication methods.
 - Data storage and transmission protection.
 - Account verification features.
5. Evaluation of Security Measures
 - How encryption works in WhatsApp.
 - Measures for account recovery and prevention of account takeover.
 - Two-step verification.
 - Backup encryption (Google Drive/iCloud).
 - Privacy settings (last seen, read receipts, profile photo visibility).
 - Spam and scam prevention.

6. Identified Strengths
 - Features that make WhatsApp secure compared to other messaging apps.
7. Identified Weaknesses & Potential Threats
 - Known vulnerabilities, phishing attempts, metadata concerns, or social engineering risks.
 - Past incidents or security breaches.
8. Recommendations
 - Suggested improvements to enhance WhatsApp's security.
9. Conclusion
 - Summarize findings and emphasize the importance of continuous security updates.
10. References
 - Use IEEE referencing style.
11. Marking Rubric

Assessment Criteria:

Criteria	Marks
Abstract and introduction clarity	10
Depth of analysis on security features	20
Critical evaluation of strengths and weaknesses	20
Practical and relevant recommendations	15
Structure, formatting, and academic writing style	15
Use of credible references	10
Overall coherence and conclusion	10

Document Preparation Guidelines:

Fonts, Page Numbering and Referencing:

- Titles: Times New Roman, Size: 14, Line spacing: 2.0, Left aligned.
- Content: Times New Roman, Size: 12, Line spacing: 1.15, Justified aligned.
- All pages should be numbered except the cover page.
- Use IEEE referencing style.

Margins:

- Normal margins (2.5 cm in all sides)

Submission

- You may submit a hard copy of your report.
- Submit your report to the department office **on or before the deadline**
- **Do not copy and paste the content of your answer from the information sources. You are required to write the answers in your own words.**
- **Copied or plagiarized (should not exceed 10%) answers will be resulted in zero marks for the assignment.**
- **Add the given Marking Rubric page to the end of your report**

Marking Rubric

Criteria	Excellent (10/9)	Good (8/7)	Satisfactory (5/4/3)	Needs Improvement (2/1/0)	Marks
Abstract & Introduction	Clear, concise, and engaging abstract; introduction provides strong background, purpose, and scope.	Mostly clear with minor gaps in purpose or scope.	Covers background but lacks clarity in purpose or scope.	Minimal or unclear introduction; abstract missing or incomplete.	/10
Depth of Analysis on Security Features	Comprehensive explanation of all major security mechanisms (encryption, authentication, backups, privacy settings) with technical accuracy.	Good explanation with some technical details missing.	General overview without much depth.	Very superficial or inaccurate explanations.	/20
Critical Evaluation of Strengths & Weaknesses	Insightful analysis with well-supported evidence; strengths and weaknesses balanced and relevant.	Some analysis with partial evidence; may lean too heavily on either strengths or weaknesses.	Limited analysis; mostly descriptive rather than evaluative.	No real evaluation; mostly vague statements.	/20
Recommendations	Practical, innovative, and realistic suggestions directly linked to identified weaknesses.	Reasonable suggestions, some lacking in practicality or detail.	Few suggestions; only loosely connected to weaknesses.	Irrelevant or no recommendations.	/15
Structure & Academic Writing	Logical flow; sections well-organized; professional tone maintained throughout; formatting perfectly followed.	Mostly logical flow; minor lapses in structure or tone.	Adequate structure; some parts disorganized or informal.	Poor structure; lacks academic style; formatting not followed.	/15
Use of Credible References	Multiple credible, up-to-date academic and industry sources; correctly cited in APA/IEEE style.	Some credible sources; minor citation errors.	Few sources; several citation errors.	No credible sources; incorrect or missing citations.	/10
Conclusion	Strong conclusion that clearly summarizes findings and reinforces importance of security.	Good conclusion with some gaps in summarizing key points.	Weak or brief conclusion; limited connection to main findings.	No conclusion or off-topic ending.	/10